Sub Code: 23CE/EE/ME/EC/CS/IT/AD/AM2T01 **MIC23**

B. Tech II Semester Regular Examinations

DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS

(Common to all Branches)

Model Paper-2

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A	**10 X 2 = 20 M**		
			CO	BTL	Marks
I.	a)	State Newton's law of cooling.	CO1	L1	2 M
	b)	Solve $xdx + ydy = \frac{a^2(xdy - ydx)}{x^2+y^2}$	CO1	L1	2 M
	c)	Find Particular Integral of $(D^2+1)y = \cos(2x-1)$	CO2	L3	2 M
	d)	Solve $(D^3+4D^2+4D+4)y = 0$	CO2	L1	2 M
	e)	Form the partial differential equation by eliminating the arbitrary function from $z = f(x^2+y^2)$.	CO3	L2	2 M
	f)	Solve $4r + 12s + 9t = 0$	CO3	L3	2 M
	g)	Define Divergence of a vector $\bar{f}$	CO4	L1	2 M
	h)	If $f(x,y,z) = 2xz^4 - x^2y$ find $\|\nabla f\|$ at(2,-2,-1)	CO4	L1	2 M
	i)	Evaluate $\oint xdy - ydx$ where C is defined by $x^2+y^2=1$	CO5	L2	2 M
	j)	State Gauss divergence theorem	CO5	L1	2 M
		PART – B	**5 X 10 = 50 M**		
		UNIT - I			
1.	a)	$Solve\ \frac{dy}{dx} + x\sin 2y = x^3\cos^2 y$	CO1	L3	5 M
	b)	If the temperature of a cup of coffee is 92^0c when freshly poured in a room having temperature 24^0c, in one minute it was cooled to 80^0c.How long a period must elapse before the temperature of the cup becomes 65^0c?	CO1	L3	5 M
		(OR)			
2.	a)	Solve $(y^4+2y)dx + (xy^3+2y^4-4x)dy = 0$	CO1	L3	5 M
	b)	If radioactive Carbon-14 has a half-life of 5750 years, what will remain of one-gram after 3000 years?	CO1	L3	5 M
		UNIT – II			
3.	a)	Solve $(D^2-4D-5)y = \cos 2x$	CO2	L3	5 M
	b)	Solve $\frac{d^4y}{dx^4} + \frac{d^2y}{dx^2} = x^3$	CO2	L3	5 M
		(OR)			
4.		Solve $(D^2+a^2)y = tanax$ by Method of Variation of Parameters	CO2	L3	10 M
		UNIT – III			
5.	a)	Form the partial differential equation by eliminating the arbitrary constants from $z = a\log(\frac{b(y-1)}{1-x})$	CO3	L2	5 M
	b)	Solve $(mz-ny)p + (nx-lz)q = ly - mx$	CO3	L3	5 M
		(OR)			

6.	a)	Solve $\frac{\partial^3 z}{\partial x^3} - 3\frac{\partial^3 z}{\partial x^2 \partial y} + 4\frac{\partial^3 z}{\partial y^3} = e^{x+2y}$	CO3	L2	5 M
	b)	Solve $(D^2 + DD' + D' - 1)z = \sin(x+2y)$	CO3	L3	5 M
UNIT – IV					
7.	a)	Find the values of the constants 'a' and 'b' so that the surface $ax^2-byz = (a+2)x$ is orthogonal to the surface $4x^2y + z^3 = 4$ at the point (1,-1,2).	CO4	L1	5 M
	b)	Find the directional derivative of $f = x^2 - y^2 + 2z^2$ at the point P (1, 2, 3) in the direction of the line PQ where Q is the point (5, 0, 4).	CO4	L1	5 M
(OR)					
8.	a)	Find the values of a, b and c such that $\bar{A}=(x+y+az)\bar{i}+(bx+2y-z)\bar{j}+(-x+cy+2z)\bar{k}$ is irrotational. Also find ϕ such that $\bar{A}=\nabla\phi$	CO4	L2	5 M
	b)	P.T $div(r^n\bar{r}) = (n+3)r^n$ .Hence S.T $r^n\bar{r}$ is solenoidal if $n+3=0$ where $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$ and $r = \lvert\bar{r}\rvert$	CO4	L2	5 M
UNIT – V					
9.	a)	If $\bar{F} = (3x^2+6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$ evaluate line integral from (0,0,0) to (1,1,1) along the path $x=t$, $y=t^2, z=t^3$.	CO5	L1	5 M
	b)	Find the work done by the force $\bar{F} = (3x^2-6yz)\hat{i} + (2y+3xz)\hat{j} + (1-4xyz^2)\hat{k}$ in moving a particle from the point (0,0 ,0) to the point (1,1,1) along the curve c $x=t$, $y=t^2$, $z=t^3$	CO5	L3	5 M
(OR)					
10.		Verify Green's theorem for $\int[(3x^2 - 8y^2)dx + (4y - 6xy)dy]$, where c is boundary of the region bounded by x = 0, y = 0, x + y = 1.	CO5	L2	10 M

***** End of the Question Paper *****